

Mathematical reference card

Every identity, constant and initializer definition the thesis depends on.
 Constants machine-checked 2026-08-15. Provenance table on the last page.

1. Setup and notation

A depth- L fully connected ReLU network of width d , no BatchNorm, zero biases at initialization:

$$z^{(\ell)} = W^{(\ell)} a^{(\ell-1)}, \quad a^{(\ell)} = \text{ReLU}(z^{(\ell)}), \quad a^{(0)} = x.$$

`nn.Linear` stores W as `(fan_out, fan_in)`, so *row* i of W is the weight vector of output neuron i , and *row centering* means $\sum_j W_{ij} = 0$.

Write $u = \text{ReLU}(z)$ with $z \sim \mathcal{N}(0, \sigma^2)$, and $R^2 = \mathbb{E}[u^2]$. Backprop error signal $\delta^{(\ell)} = \partial \mathcal{L} / \partial z^{(\ell)}$, with $\partial \mathcal{L} / \partial W^{(\ell)} = \delta^{(\ell)} (a^{(\ell-1)})^\top$ — so a **parameter** gradient is the product of a forward and a backward quantity. This is why the two directions cannot be tuned independently.

Per-layer gains:

$$g_{\text{fwd}} = \frac{\text{rms}(a^{(\ell)})}{\text{rms}(a^{(\ell-1)})}, \quad g_{\text{bwd}} = \frac{\text{rms}(\delta^{(\ell)})}{\text{rms}(\delta^{(\ell+1)})}.$$

2. Half-Gaussian moments — the atoms

Everything below is built from three facts about $u = \text{ReLU}(z)$, $z \sim \mathcal{N}(0, \sigma^2)$:

$$\mathbb{E}[u] = \frac{\sigma}{\sqrt{2\pi}}, \quad \mathbb{E}[u^2] = \frac{1}{2}\sigma^2, \quad \text{Var}(u) = \sigma^2 \left(\frac{1}{2} - \frac{1}{2\pi} \right). \quad (1)$$

Two ratios follow, and they are the whole story:

$\frac{\mathbb{E}[u]}{\text{rms}(u)} = \frac{1}{\sqrt{\pi}} \approx 0.5642$	(the DC component, relative to scale) (2)
$\frac{\text{Var}(u)}{\mathbb{E}[u^2]} = \frac{\pi - 1}{\pi} \approx 0.6817$	(the <i>centering ratio</i>) (3)

Why the centering ratio matters. A row-centered W annihilates the constant component of its input: if $\sum_j W_{ij} = 0$ then $W\mathbf{1} = 0$, so $z_i = \sum_j W_{ij}(a_j - \bar{a})$. The layer therefore sees $\text{Var}(a)$, not $\mathbb{E}[a^2]$ — it is *blind to the DC*. An uncentered (He) layer sees $\mathbb{E}[a^2]$. The ratio between what the two see is exactly $(\pi - 1)/\pi$.

3. The arc-cosine kernel and geometric collapse

For z_k, z_m jointly Gaussian with correlation ρ and equal variance, write $\alpha = \arccos \rho$. Then

$$\chi(\rho) = \frac{\mathbb{E}[\text{ReLU}(z_k) \text{ReLU}(z_m)]}{\mathbb{E}[\text{ReLU}(z)^2]} = \frac{\sin \alpha + (\pi - \alpha) \cos \alpha}{\pi} \quad (4)$$

with $\chi(1) = 1$, $\chi(0) = 1/\pi \approx 0.3183$, $\chi(-1) = 0$. Per-sample norms cancel, so χ depends only on the angle.

Two fixed points — the two collapse modes

- **He (uncentered).** $\chi(\rho) > \rho$ on $[0, 1)$, so $\rho_\ell \uparrow 1$: every pair of inputs becomes parallel. Measured: mean pairwise cosine 0.993 at $L = 60$.
- **Row-centered.** The DC is removed each layer, and the iteration lands on a *non-degenerate* fixed point $\rho^* \approx 0.32$, i.e. $\approx 71^\circ$ — stable from $L = 20$ to $L = 60$. Collapse is avoided; §5 gives the price.

Local expansion at $\rho = 1$ (the rate)

Since $\chi'(1) = 1$, the map is tangent to the identity and convergence is *algebraic*. With $u = 1 - \rho$:

$$1 - \chi(\rho) = u - \frac{2\sqrt{2}}{3\pi} u^{3/2} + O(u^2), \quad c_\chi := \frac{2\sqrt{2}}{3\pi} \approx 0.30011 \quad (5)$$

Solving $u_\ell = u_{\ell-1} - c_\chi u_{\ell-1}^{3/2}$:

$$\varepsilon_\ell = 1 - \rho_\ell \approx \frac{4}{c_\chi^2} \ell^{-2} = \frac{9\pi^2}{2} \ell^{-2} \approx \frac{44.413}{\ell^2}, \quad \Delta_\ell = \frac{\arccos(1 - \varepsilon_\ell)}{\pi} \approx \frac{3}{\ell} \quad (6)$$

$4/c_\chi^2 = 9\pi^2/2$ is an *exact* identity. The leading constant is independent of ρ_0 .

4. The gain-coupling lock

$$\frac{g_{\text{fwd}}}{g_{\text{bwd}}} = r = \sqrt{\frac{\pi-1}{\pi}} \approx 0.82565 \quad (\text{independent of the weight variance}) \quad (7)$$

The forward pass of a row-centered layer sees $\text{Var}(a)$; the backward pass sees $\mathbb{E}[a^2]$ (the error signal carries no DC constraint). Their ratio is the centering ratio, and its square root is the ratio of *gains*. Scaling the variance moves both gains together and cannot change their ratio — hence a *lock*, not a tunable trade-off.

choice	$\text{Var}(W) \cdot d$	g_{fwd}	g_{bwd}	realized by
backward-balanced	2	0.826	1.000	<code>row_centered_he</code>
product-balanced	$2/r \approx 2.4223$	$\sqrt{r} \approx 0.909$	$1/\sqrt{r} \approx 1.100$	<code>row_centered_product_balanced</code>
forward-balanced	$2\pi/(\pi-1) \approx 2.9339$	1.000	$1/r \approx 1.211$	<code>row_centered_forward_balanced</code>

Over L layers the uncorrected cost is r^L forward or r^{-L} backward. At $L = 100$: $r^{100} \approx 4.78 \times 10^{-9}$ and $r^{-100} \approx 2.09 \times 10^8$.

5. Post-ReLU DC removal, and the duality

Modify the forward pass (*not* the weights): $a \leftarrow \text{ReLU}(z) - c \text{rms}(a)$.

Derivation of $G(c)$

The gain is the rms of the shifted activation relative to the unshifted one, $G(c) = \text{rms}(u - cR) / \text{rms}(u)$ with $u = \text{ReLU}(z)$ and $R = \text{rms}(u)$, so $R^2 = \mathbb{E}[u^2]$. Expand the square and substitute the half-Gaussian mean $\mathbb{E}[u] = R/\sqrt{\pi}$ from (2):

$$\mathbb{E}[(u - cR)^2] = \mathbb{E}[u^2] - 2cR\mathbb{E}[u] + c^2R^2 = R^2 - 2cR \cdot \frac{R}{\sqrt{\pi}} + c^2R^2 = R^2 \left(1 - \frac{2c}{\sqrt{\pi}} + c^2\right).$$

Divide by R^2 and take the square root:

$$G(c) = \sqrt{1 - \frac{2c}{\sqrt{\pi}} + c^2}, \quad G\left(\frac{1}{\sqrt{\pi}}\right) = \sqrt{1 - \frac{1}{\pi}} = \sqrt{\frac{\pi-1}{\pi}} = r \quad (8)$$

Minimising: $\frac{d}{dc}(c^2 - 2c/\sqrt{\pi}) = 2c - 2/\sqrt{\pi} = 0$ gives $c = 1/\sqrt{\pi}$ — and $cR = R/\sqrt{\pi} = \mathbb{E}[u]$ there, so the minimising shift removes *exactly the mean*. At the minimum

$$G^2\left(\frac{1}{\sqrt{\pi}}\right) = 1 - \frac{1}{\pi} = \frac{\mathbb{E}[u^2] - \mathbb{E}[u]^2}{\mathbb{E}[u^2]} = \frac{\text{Var}(u)}{\mathbb{E}[u^2]},$$

which is the centering ratio (3) — the same quantity row centering pays, hence $G(1/\sqrt{\pi}) = r$ with no coincidence involved. Also $G(c) < 1$ for all $c \in (0, 2/\sqrt{\pi})$, $2/\sqrt{\pi} \approx 1.1284$.

Duality (why this is not a new family)

$$W(a - c\mathbf{1}) = Wa - c(W\mathbf{1}) \implies W\mathbf{1} = 0 \Rightarrow \text{the shift is exactly a no-op.}$$

Row centering kills the DC *on the way in*; the shift kills it *on the way out*. They are one intervention, and **both pay the identical constant r** . The lock is therefore a property of DC removal itself, not of row centering.

Cosine recursion under the shift

With $A(c) = c^2 - 2c/\sqrt{\pi}$ (so $G^2 = 1 + A$):

$$\rho_{\ell+1} = \frac{\chi(\rho_\ell) + A}{1 + A}.$$

$\rho = 1$ stays a fixed point but its multiplier becomes $1/(1 + A) > 1$ — *repelling*, which is precisely why the shift attacks collapse. The attracting fixed point solves $\Phi(\rho^*) = -A$ with $\Phi(\rho) = (\chi(\rho) - \rho)/(1 - \rho)$, decreasing on $[0, 1)$ from $\Phi(0) = 1/\pi$. Since $\max_c |A| = 1/\pi$ attained at $c = 1/\sqrt{\pi}$, theory predicts $\rho^* = 0$ there and $\rho^* > 0$ on both sides.

The cost is transferable, not prohibitive

Boosting the weights by $1/G(c)$ restores $g_{\text{fwd}} \approx 1$ and drives $g_{\text{bwd}} \rightarrow 1/r$, with $g_{\text{fwd}}/g_{\text{bwd}}$ invariant under the boost — the signature of a lock. So (i) and (ii) are not *provably incompatible*; the cost simply moves between the two directions.

6. grad_rescale: the backward-only correction

$\mathcal{R}_r$ is the identity in the forward pass and multiplies the gradient by r in the backward pass, inserted after every hidden ReLU. It cancels the forward-balanced $g_{\text{bwd}} = 1/r$ exactly, and is equivalent to a geometric per-layer learning rate $r^{L-\ell}$.

Reach law. A trainable layer at position ℓ picks up r raised to the number of rescale operations *between it and the loss*. With the rest of the network frozen, autograd stops at the first fully-frozen upstream chain, so the reach is wildly asymmetric:

$$\text{tail window } \{f_{L-1}, f_L\} : r^1, r^2 = 0.826, 0.682 \quad \text{front window } \{f_1, f_2\} : r^L, r^{L-1} \approx 5 \times 10^{-9}.$$

Consequence: at the tail the (init $\times$ rescale) 2×2 collapses to a 1×2 — only the initialization matters. Because $\mathcal{R}_r$ cannot touch the forward pass, any explanation that runs through activations must be an *initialization* story.

7. Dying neurons

A unit is *dataset-dead* at layer ℓ if $z_i^{(\ell)}(x_k) < 0$ for every x_k in the dataset; its entire row of $\partial\mathcal{L}/\partial W^{(\ell)}$ is then zero.

Let $A = \{\text{all } N \text{ signs agree}\}$ and $A_- = \{\text{all negative}\} = \{\text{dead}\}$. Since $A = A_+ \uplus A_-$ and $w \mapsto -w$ gives $\mathbb{P}(A_+) = \mathbb{P}(A_-)$:

$$\boxed{\mathbb{P}[\text{dead}] = \frac{1}{2} \mathbb{P}(A) \quad (\text{exact — no limit required})} \quad (9)$$

As $\rho \rightarrow 1$ the activations become collinear, $\mathbb{P}(A) \rightarrow 1$, and $\mathbb{P}[\text{dead}] \rightarrow \frac{1}{2}$.

Pairwise (union) bound — vacuous in practice

$\mathbb{P}(A) \geq 1 - \binom{N}{2} \Delta_\ell$ needs $\ell \gtrsim \frac{3}{2} N(N-1)$; for $N = 60,000$ that is depth 5.4×10^9 .

Slepian bound — the version that bites

$\mathbb{P}[\forall k : Y_k > 0]$ is monotone in every pairwise correlation, so the equicorrelated configuration at the floor $\rho = 1 - \varepsilon$ is *extremal*. Writing $Y_k = \sqrt{1-\varepsilon} G + \sqrt{\varepsilon} U_k$ with G, U_k i.i.d. $\mathcal{N}(0, 1)$:

$$\mathbb{P}(A) = 2 \int \phi(g) \Phi(\lambda g)^N dg, \quad \lambda = \sqrt{\frac{1-\varepsilon}{\varepsilon}}, \quad \mathbb{P}(A) \geq 1 - C \sqrt{\varepsilon \log N}.$$

The N -dependence is *logarithmic*, not quadratic: non-vacuous at $\ell > 43$ for $N = 60,000$.

Measured

He, width 500, $N = 512$ probe samples (this quantity is *not* scale-free in N): dead fraction 0.342/0.400/0.476 at $L = 30/60/100$. At $L = 60$ the Slepian bound gives $\mathbb{P}[\text{dead}] \geq 0.3881$ against a measured 0.4000 — tight to 3%.

8. Initializer catalogue

All rows: draw $W_{ij} \sim \mathcal{N}(0, \sigma^2)$ with σ^2 as given, biases zero. “center” = subtract the row mean; “rescale” = set each row’s empirical std to the target. Row centering alone reduces per-element variance by $(1 - 1/d)$.

name	definition ($\sigma^2 d$, then structure)	row sum 0	lock side
<i>Baselines</i>			
he	2	no	—
xavier	$2d/(\text{fan}_{\text{in}} + \text{fan}_{\text{out}})$	no	—
uniform_he	$U(-a, a)$, $a^2/3 = 2/d$	no	—
orthogonal	Q from QR, gain configurable	no	—
orthogonal_he	$Q \cdot \sqrt{2}$	no	—
orthogonal_tuned	$Q \cdot \sqrt{1.65}$	no	—
<i>Row-centered family</i>			
row_centered_he	2, center	yes	bwd-bal. ($g_{\text{fwd}} = r$)
row_centered_he_var_adj	2, center, rescale	yes	—
row_centered_final	1.65, center, rescale	yes	tuned to $g_{\text{fwd}} \approx 1$
row_centered_forward_balanced	$2\pi/(\pi - 1) \approx 2.9339$, center, rescale	yes	fwd-bal. ($g_{\text{bwd}} = 1/r$)
row_centered_product_balanced	$2/r \approx 2.4223$, center, rescale	yes	$g_{\text{fwd}} g_{\text{bwd}} = 1$
row_centered_layer_balanced	per-layer $\sigma^2(\ell)$ schedule, η	yes	gradient uniformity
row_centered_layer_balanced_he_base	as above, He base	yes	backward-aware
row_centered_layer_balanced_product_base	as above, product base	yes	V2
<i>Interpolating / experimental</i>			
partial_centered_he	2, center by α , rescale	partial	α -interpolation
centered_with_dc_he	2, center, rescale, $+\delta\sqrt{2/d}$	nearly	breaks the constraint
kernel_preserving	optimizer minimising kernel distortion	no	—
custom_variance	free σ^2 , free mean	no	—
<i>Forward-pass modifiers (not registry initializers)</i>			
relu_shift=c	$a \leftarrow \text{ReLU}(z) - c$ rms(a), He weights	—	$g_{\text{fwd}} = G(c)$
grad_rescale=r	identity fwd, $\times r$ bwd	—	cancels g_{bwd}
trainable_layers	requires_grad=False elsewhere	—	—

How the families relate. row_centered_forward_balanced is row_centered_he with rows rescaled by $\approx 1/r$: same random draw, same statistical geometry (mean pairwise cosine ≈ 0.32 at every depth for both), differing only in scale by $r^{-\ell}$. And by §5, relu_shift on He weights is the activation-space image of the same DC removal. *Three names, one mechanism.*

9. Constants

symbol	meaning	value
$r = \sqrt{(\pi - 1)/\pi}$	the gain-coupling constant	0.8256452712
$1/r$	backward gain, forward-balanced	1.2111738960
$r^2 = (\pi - 1)/\pi$	centering ratio $\text{Var}(u)/\mathbb{E}[u^2]$	0.6816901138
$1/\sqrt{\pi}$	DC of a rectified Gaussian, $\mathbb{E}[u]/\text{rms}(u)$	0.5641895835
$1/\sqrt{2\pi}$	$\mathbb{E}[u]/\sigma$	0.3989422804
$2\pi/(\pi - 1)$	forward-balanced $\sigma^2 d$	2.9338844140
$2/r$	product-balanced $\sigma^2 d$	2.4223477920
$2/\sqrt{\pi}$	upper root of $G(c) = 1$	1.1283791671
$c_\chi = 2\sqrt{2}/(3\pi)$	cubic coefficient in $1 - \chi$	0.3001054387
$4/c_\chi^2 = 9\pi^2/2$	constant in ε_ℓ	44.4132198
$\chi(0) = 1/\pi$	one-layer decorrelation floor	0.3183098862
r^{100}	forward decay, $L = 100$	4.779×10^{-9}
r^{-100}	backward blow-up, $L = 100$	2.092×10^8
$\rho_{\text{row-centered}}^*$	non-degenerate fixed point ($\approx 71^\circ$)	≈ 0.32

10. Provenance

result	where it is established / measured
Half-Gaussian moments, centering ratio	ch3_gradient_trap.tex (Lemma, Proposition)
DC blindness	ch3_gradient_trap.tex (Proposition)
Gain-coupling lock $g_{\text{fwd}}/g_{\text{bwd}} = r$	ch3_gradient_trap.tex, prop:fb_ratio
Arc-cosine kernel, angle contraction	ch1_framework.tex (Theorem, Proposition)
Two fixed points, the 71° result	ch7_angle_map.tex
$G(c)$, duality, lock generalised to DC removal	cluster/11_relu_shift/README.md; <i>not yet in any chapter</i>
grad_rescale reach law	reports/results/recipe_decomposition_funnel.json
ε_ℓ , Δ_ℓ , $\mathbb{P}[\text{dead}] = \frac{1}{2}\mathbb{P}(A)$	docs/scratch/proofs/dying_neurons_clean_proof.tex
Slepian bound	docs/scratch/proofs/oracle_spotcheck_addendum.md — sketch, needs proving
Dead fractions, cosine, k -NN measurements	reports/results/relu_shift_geometry_screen_*.json
Empirical outcomes of all campaigns	docs/reports/2026-08-15_campaign10_followup_synthesis.md

All numerical constants were recomputed from their closed forms and checked against simulation before this card was written. The only item on it that is *not* yet a proof is the Slepian step in §7.